GUANHUA DING

Software Engineer · Backend (Java / TypeScript) · AI-Assisted Delivery

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Backend software engineer with 7 years of experience building production APIs, event-driven architectures, and distributed services. Over the past three years across Leonardo.Ai and Canva, built end-to-end generative-AI pipelines, cross-platform SSO authentication, and enterprise admin backends using Java and TypeScript/Node.js. Experienced in integrating AI coding agents (Claude Code, Cursor) into daily engineering workflows backed by strict specification design, automated testing, and code review.

EXPERIENCE

Canva | Software Engineer (Backend)

Jan 2024 – Present

Leonardo.Ai (Jan 2024 – May 2026), acquired by Canva — generative-AI image, video, and design platform

- **Enterprise & Education Admin (Canva):** Engineered and maintained Java backend services powering Team Admin capabilities for enterprise and education organizations, focusing on large-scale user account consolidation and cross-account asset migration.
- **Migration Reliability & Speed:** Re-architected asynchronous asset transfer workflows and batch processing logic, significantly reducing migration execution time and lowering failure rates on large organizational datasets.
- **Generative-AI Pipeline (Leonardo.Ai):** Built the core image-generation backend pipeline connecting web and iOS clients to AI inference workers using TypeScript, Node.js (Nest.js), PostgreSQL, and AWS Lambda.
- **Async Orchestration & Webhooks:** Designed event-driven state synchronization using Hasura GraphQL and webhooks to handle long-running AI generation callbacks reliably under heavy concurrent load.
- **Cross-Platform Auth & SSO:** Delivered authentication services and implemented Single Sign-On (SSO) integration between Leonardo.Ai and Canva following the acquisition, enabling unified user identity across both platforms.
- **AI-Augmented Engineering:** Integrated Claude Code and Cursor into daily development using spec-first prompting and strict automated test gates to accelerate service scaffolding, complex refactors, and unit/integration test coverage.

Go1 | Software Developer

Oct 2021 – Jan 2024

B2B learning and compliance platform

- **Third-Party Integrations:** Designed and shipped backend services in TypeScript, Node.js, and Nest.js to integrate the core learning platform with external third-party LMS and enterprise HR ecosystems.
- **Event-Driven & Webhook Architecture:** Built and scaled serverless microservices on AWS and Azure to process high-volume outbound webhooks and asynchronous domain events, ensuring reliable data synchronization across partner systems.
- **Full-Lifecycle Delivery:** Contributed to React frontend portals, maintained CI/CD pipelines, and upheld service reliability through comprehensive REST API integration tests, unit testing, and technical documentation.

Digistorm | Software Engineer **May 2019 – Oct 2021**

CRM web applications for K-12 schools

- **Full-Stack CRM Development:** Developed and maintained core K-12 school enrollment and CRM web applications across the full stack using Vue.js, TypeScript, and PHP (Laravel).
- **API & Data Layer Optimization:** Built third-party API integrations to sync student and parent records with school management systems, and optimized PostgreSQL and Elasticsearch queries to improve data retrieval responsiveness.

SKILLS

Languages	TypeScript, JavaScript, Java, SQL, PHP
Backend	Node.js, Nest.js, Spring Boot, GraphQL (Hasura), REST APIs, Webhooks, Event-Driven Microservices, PostgreSQL, Elasticsearch
Cloud & DevOps	AWS (Lambda, S3, SQS/SNS), Azure, Docker, CI/CD
Frontend	React, Next.js, Vue.js, Tailwind CSS
AI Workflow	Claude Code, Cursor, GitHub Copilot, Spec-First Prompting, MCP, Agent-Assisted Testing & Refactoring

EDUCATION

General Assembly, Sydney | Web Development Immersive **Dec 2018 – Mar 2019**

Queensland University of Technology (QUT) | Diploma of Business **2013 – 2014**